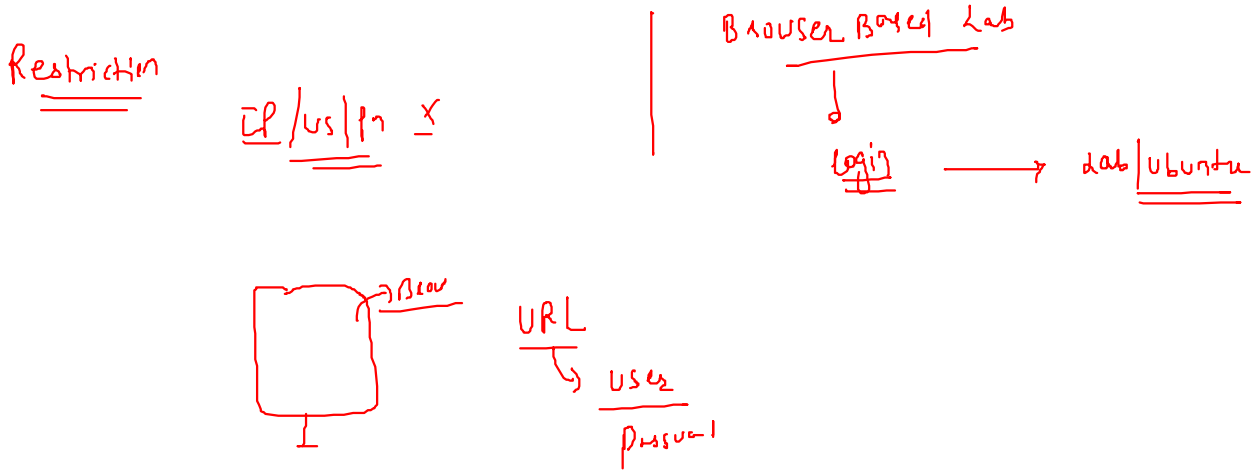


4 things

1. Theory 20% and practical 80%
2. Interactive
3. Notes :-
 - google drive - folder
 - Day wise -- word file --
 - commands - Diagram
 - tasks

4. Labs



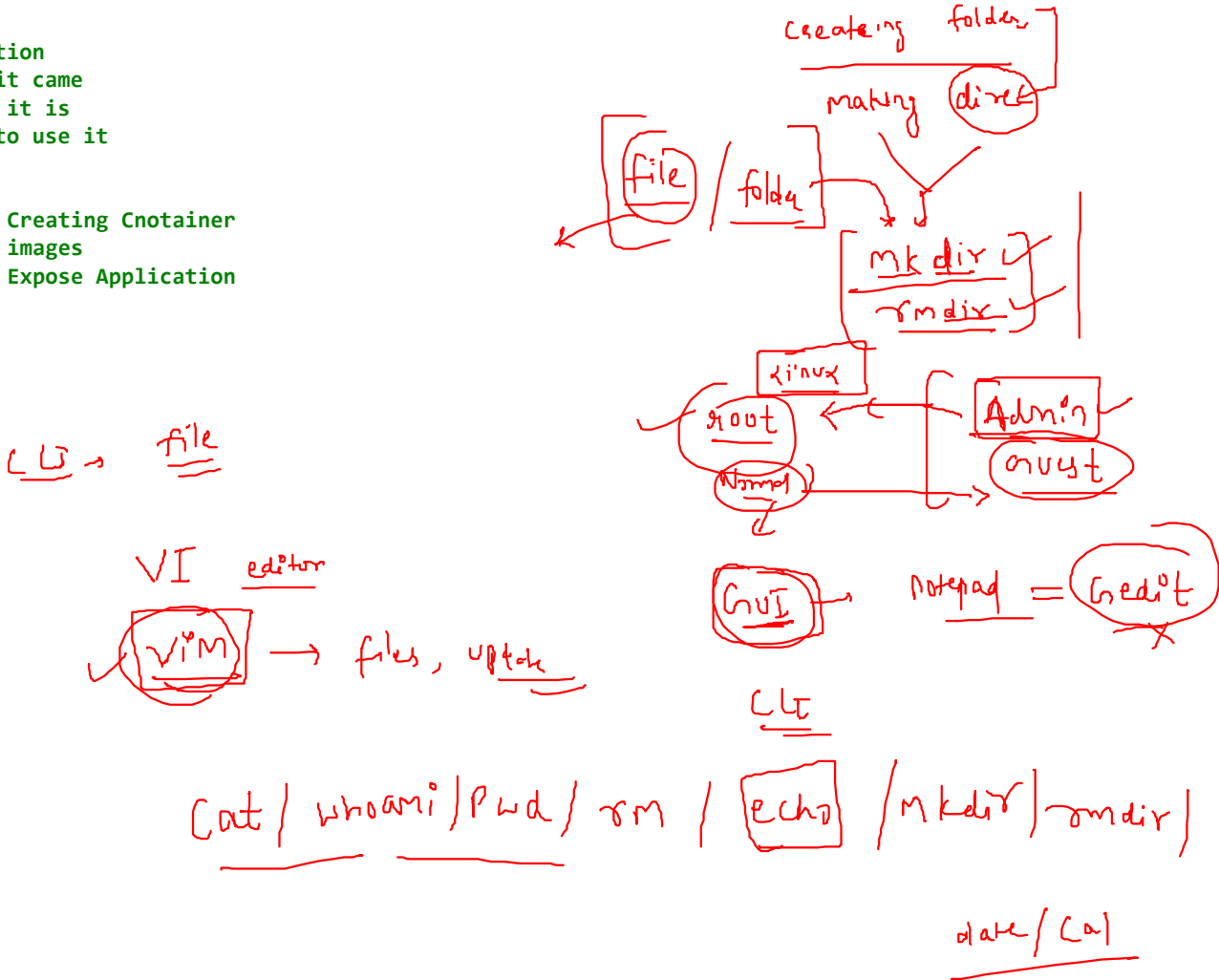
Today's Agenda :-

1. Basic Linux
2. Containerization
 - a. why it came
 - b. what it is
 - c. how to use it

Docker

Creating Container images

Expose Application



Phase-1 → 1990's

① 1. Physical Server - 40 Days ✓

2. Build the code - 40 Days

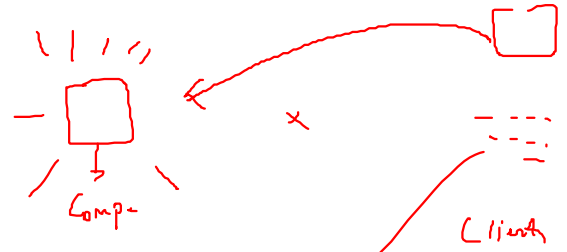
3. Install OS - Windows - 12 Hours

4. Install/ Configure webserver { apache , nginx , tomcat , } - 2 hours

5. Copy your code correction location

6. start the service of apache

7. configure DNS -- xyz.com



✓ 21 days ✓

2 M.A

URL

Alk



Phase 2 -- 2000's -- virtualization -- vmware , virtualbox, hyperv,

1. Physical Server - 10 Days

2. Coding is done in 10 Days

3. Install OS -- 3 hours

4. Install Virtualization -- 6 hours

5. Createing VM -- 6 hours

6. Install Apache in VM

7. copy code

8. Configure DNS

12 days

1hr



Solomen hykes

2006

Phase 3 - 2013's Conterinization - docker , buildah , runc , podman , crio containerd

1. VM → 1 mins

2. docker → 10 sec on

↓
① creat container →

(X)

← 1s

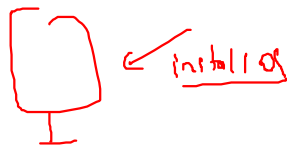
1s

Yes

→
10 Cont

→ 10 sec,

without OS



Bootable CD

①

iso file

②

PD



1GB

1GB → 2GB

4GB bind to

software

boot

driver all

docker hub



Apache

① Custom image



OS +

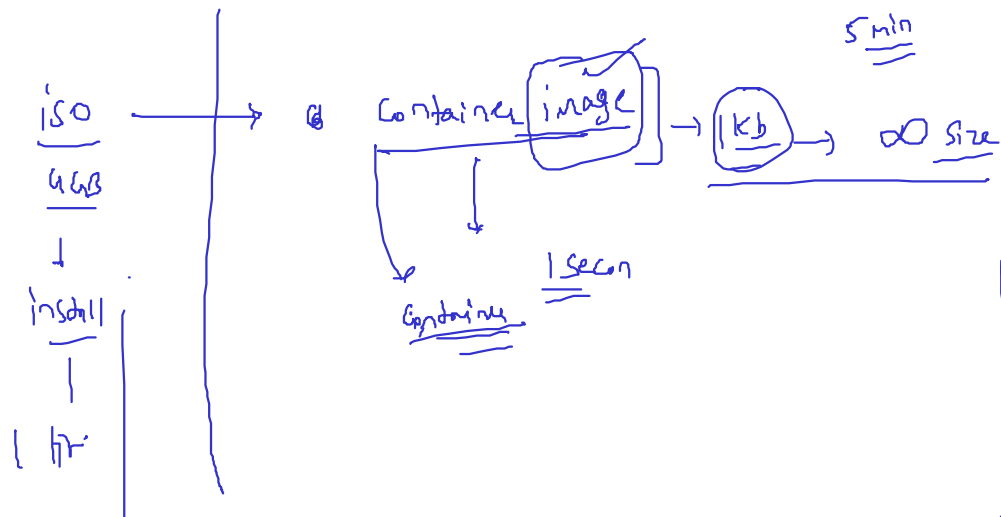
docker image
Container

docker =

run

Containers

create



VM

4GB

20GB

④

Virtualization

6GB

VM2

4GB

VM3

kernel

VMWARE

OS

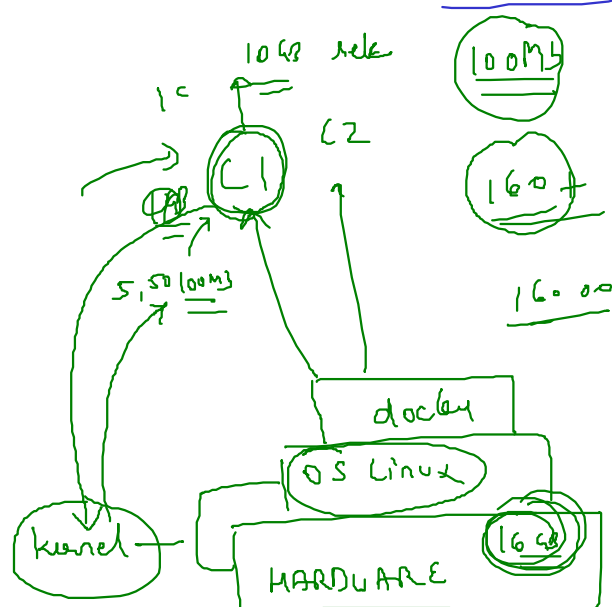
HARDWARE

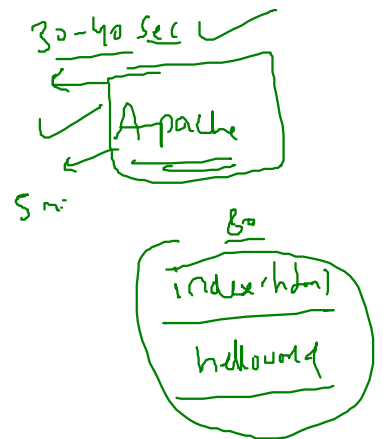
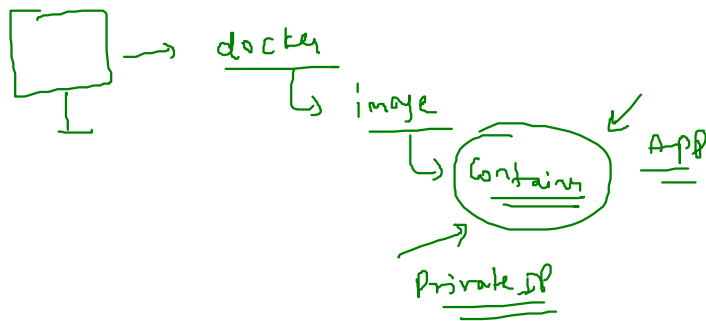
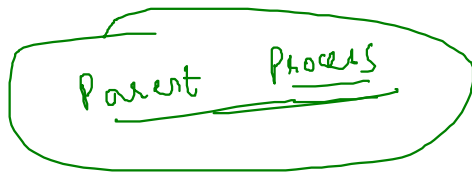
16GB

200GB

15VM

Containerization



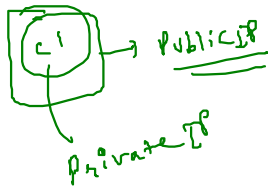


Pomcat

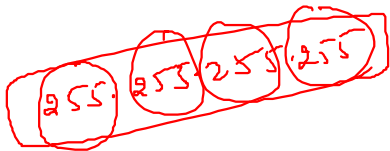
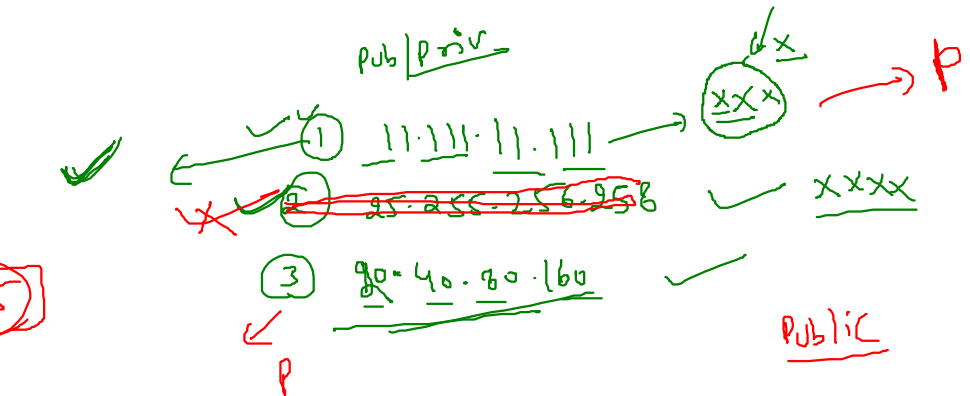
JAR / WAR

- 1) install correct Package [apache ✓
- 2) Config/ index.html
- 3) Start server

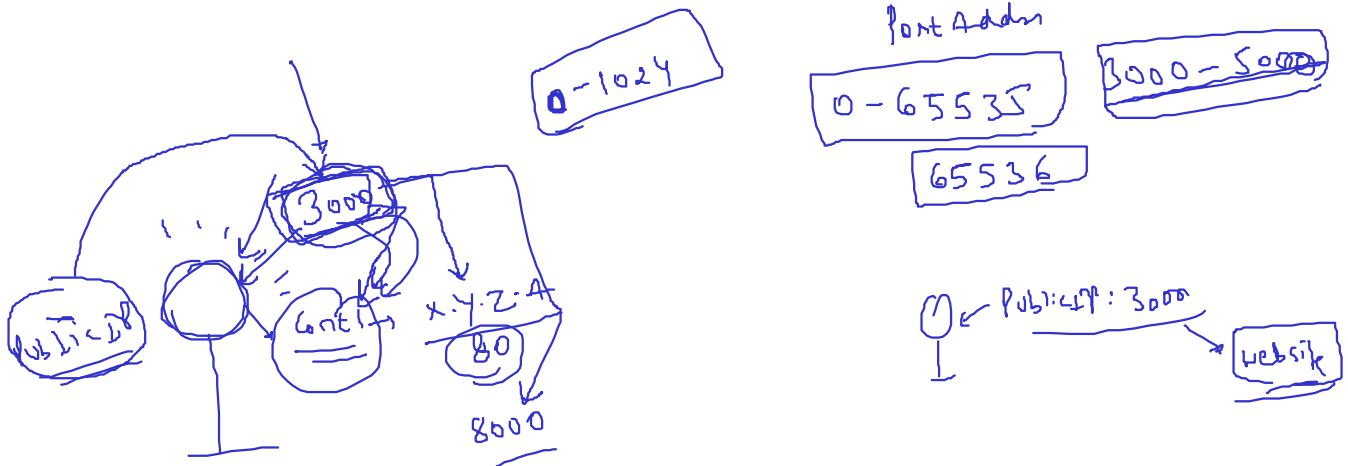
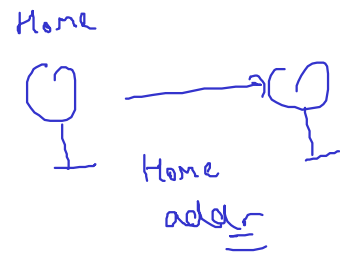
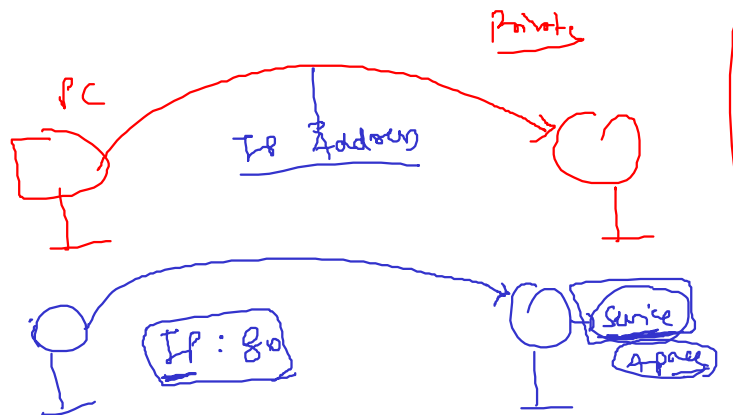
30-4054



Port forwarding



- Private →
- ① 10.X.X.X ✓ Private
 - ② 192.168.X.X → Drinck
 - ③ 172.16.X.X to 172.31.X.X



Custom image →

Dockerfile

Instructions

① FROM



② RUN

③ RUN / COPY

④

hhttpd -Dfore

CMD

ENTRYPOINT

⊗

① Choose image

② Create Container ✓

③ Changes

Installation

Apache

creating file

Start service

Parent process